



Bacteria and Bacterial Diseases

Rapid impedance-based Antimicrobial Susceptibility Testing (iFAST) of *Enterobacterales* in urinary tract infections

Lucy J. Bock^{a,*}, Daniel C. Spencer^b, Bethany K. Martin^{a,b}, Caitlin N. Daniels^a, Xena Dyball^a, Charlotte K. Hind^a, Matthew E. Wand^a, Collette E. Allen^a, Robert C. Read^c, H. Morgan^b, J. Mark Sutton^{a,d,**}

^a Antimicrobial Discovery, Development and Diagnostics, Countermeasures Development, Evaluation and Preparedness, UK Health Security Agency, Porton Down, Salisbury, Wiltshire SP4 0JG, UK

^b School of Electronics and Computer Science, and Institute for Life Science, University of Southampton, Southampton, Hampshire SO17 1BJ, UK

^c NIHR Southampton Biomedical Research Centre, School of Clinical and Experimental Science, and Institute for Life Science, University of Southampton, University Hospital Southampton, Tremona Road, Southampton SO16 6YD, UK

^d Institute of Pharmaceutical Sciences, School of Cancer & Pharmaceutical Sciences, King's College London, SE1 9NQ, UK

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SUMMARY

Objectives: Evidence-based antibiotic prescribing for urinary tract infections (UTIs) would increase treatment success and improve antibiotic stewardship. Current antimicrobial susceptibility tests (AST) are time-consuming. A novel phenotypic impedance-based Fast AST (iFAST) measures changes in the electrical phenotype of single bacteria in response to antibiotic exposure. Suitability of this technology for UTI causing bacteria was investigated.

Methods: Fifty-eight strains of *Escherichia coli* and *Klebsiella pneumoniae* were exposed to EUCAST breakpoint concentrations of UTI antibiotics. Following a two-hour exposure, the % cell count compared to unexposed control populations were compared and susceptibility deduced. Results were compared to gold standard broth microdilution (BMD) AST results. Susceptibility thresholds were clinically evaluated. Strain-antibiotic combinations with a minimum inhibitory concentration (MIC) on or one doubling dilution above the breakpoint were exposed to doubling dilutions of antibiotics and measured on iFAST to determine an electrical MIC.

Results: 100% correlation was obtained for all eight antibiotics against laboratory strains, when allowing for the inherent 2-fold variability of the BMD MIC measurement, within a five-hour test. Clinical evaluation showed concordance in at least 74 out of 80 tests. Electrical MICs showed broad equivalence with classical MICs.

Conclusions: iFAST has potential as an accurate and rapid AST for UTI causing *Enterobacterales*.

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Introduction

Urinary tract infections (UTIs) are one of the most common infections worldwide, with over 400 million cases estimated in 2019.¹ Prompt treatment of UTIs is an important component of antibiotic prescribing² and if left untreated or treated with ineffective antibiotics, UTIs can develop into bacteraemia. In fact, nearly half of all

Escherichia coli bacteraemia in the UK originate from a UTI.³ Prescribing effective antibiotics for each UTI as soon as a patient presents, rather than empirical prescribing, would reduce morbidity, mortality and cost of ineffective treatment.⁴ Antimicrobial resistance (AMR) is already a significant contributor to the global burden of disease⁵ and likely to increase,⁶ further emphasising the need to reduce incorrect prescribing.

Antimicrobial susceptibility tests (AST) on urinary pathogens in diagnostic laboratories can take from 40–72 h from sample reception to susceptibility read-out. This delay leads to empirical antibiotic prescribing,⁷ bringing with it the risk of ineffective treatment and poor antibiotic stewardship. A range of automated systems have been developed for use in hospitals to help reduce the time to results and automate various aspects of the process. Examples include the

* Corresponding author.

** Corresponding author at: Antimicrobial Discovery, Development and Diagnostics, Countermeasures Development, Evaluation and Preparedness, UK Health Security Agency, Porton Down, Salisbury, Wiltshire SP4 0JG, UK.

E-mail addresses: lucy.boock@ukhsa.gov.uk (L.J. Bock), mark.sutton@ukhsa.gov.uk (J.M. Sutton).

MAST Uri System,^{8,9} which automates plating onto pre-poured antibiotic agar and uses image analysis and software to determine susceptibility after 20 h of exposure, or the Thermo Fisher Sensititre™ Gram Negative Urine Plate or the Beckman Coulter MicroScan NMDRM1,¹⁰ for high-throughput broth microdilution. These methods are principally phenotypic susceptibility tests, which measure the growth of the bacteria, either on agar or in liquid media, and use different methods to accelerate the interpretation of changes in growth rate or bacterial survival. They rely on bacteria growing to a measurable level, which usually requires 18 h, though a rapid disc diffusion AST reading at 4, 6 or 8 h post inoculation is available from EUCAST. However, this is often difficult to implement in the laboratory workflow and is currently only available for positive blood cultures.

Recently, the PA-100 AST System from Sysmex Astrego was awarded the UK's Longitude Prize for its rapid UTI test, currently being evaluated as a method for assessing bacteriuria and phenotypic susceptibility.¹¹ The test tracks the growth profile of a defined set of bacterial species *E. coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, *Enterococcus faecalis* and *Staphylococcus saprophyticus* extracted directly from urine and exposed to one of amoxicillin/clavulanic acid, ciprofloxacin, fosfomycin, nitrofurantoin and trimethoprim. The authors state that the 45-minute test improves the optimal treatment recommendation compared to routine clinical methods.¹¹

Other novel ASTs include the use of viability markers, for example non-fluorescent resazurin is reduced to the strongly fluorescent resorufin by growing bacteria. This can then be detected using various optical,¹² electrochemical¹³ or other methods. Companies developing fast ASTs include Resistell™ who measure the nanomotion of bacteria¹⁴; Innovative Dx who measure bacterial growth in a proprietary media using fluorescent dyes and imaging¹⁵; Lifescale (Affinity Biosensors) using microfluidic sensors to count and measure the mass of individual organisms¹⁶; and Microplate Dx who use electrochemical impedance spectroscopy to detect bacterial growth.¹⁷ Although there are many technologies in development,¹⁸ there remain issues around expensive equipment and consumables or the need for specific reagents such as fluorescent dyes.

Genotypic tests to detect resistance markers are also available.¹⁹ However, these are limited to known resistance markers and cannot detect phenotypic resistance, such as efflux pump upregulation. In addition, genotypic tests are costly and therefore not routinely used in high throughput urine analysis.

Single cell impedance cytometry is a relatively simple label-free technique that can rapidly measure the effect of antimicrobials on many individual bacterial cells.²⁰ Following a short exposure to antimicrobials, the electrical properties of several thousand single organisms are measured in a microfluidic device in a time-frame of two minutes. Susceptible bacteria show clear differences in electrical impedance within as little as 30 min of exposure due to a combination of changes in cell volume and the electrical properties (conductivity/permeability) of the cell wall/membrane. Comparison of the electrical properties of exposed and unexposed populations provides a simple way to determine the susceptible or resistant profile of a range of bacterial species to a number of antibiotics. This reagent free technology is termed impedance-based Fast Antimicrobial Susceptibility Test (iFAST).

In this manuscript, iFAST was used to examine the susceptibility of 58 bacterial strains with results compared to the current gold standard broth microdilution (BMD) AST, as described by the ISO standard ISO 20776-1:2019²¹ and aligned with EUCAST methodology. The study focussed on *E. coli* and *K. pneumoniae*, the two most common causes of urinary tract infection, and the pathogens most associated with high levels of antimicrobial resistance.^{22,23} The bacteria were exposed to a panel of eight different antibiotics recommended for frontline treatment and management of patients with UTI. Results demonstrate that the iFAST has good concordance

with susceptibility determined by broth microdilution (BMD), either at a single breakpoint concentration or as part of an electrical Minimum Inhibitory Concentration (eMIC) determination.

Materials and methods

Bacterial strains and reagents

The strains used in this study are listed in Table S1.^{24–28} Strains were grown on cation-adjusted Mueller Hinton (MH2) or cystine-lactose-electrolyte-deficient (CLED) agar plates (EO Laboratories) or in MH2 broth (MHB, Sigma) at 37°C for 20 h. Strains remained on plates for a maximum of 48 h. The following antibiotics from Sigma were dissolved in water: potassium clavulate (with amoxicillin, AMC), ceftazidime hydrate (CAZ), cefalexin (CEX), gentamicin sulfate (GEN); in water with acetic acid: ciprofloxacin (CIP) and trimethoprim (TMP); in DMSO: amoxicillin (AMX); or in DMF: nitrofurantoin (NIT) and then further diluted in water before adding to the cells in MH2 broth. Dissolved antibiotics were stored at 4°C for a maximum of seven days before use. 2 µm carboxylated polystyrene beads (Polysciences) were added to 0.9% saline for normalisation. All media, saline and diluents were autoclaved and filtered through a 0.22 µm filter before use.

Impedance cytometry system

The experimental system has been described in detail elsewhere.²⁰ A miniature cytometer is based around a microfluidic chip that measures the electrical properties of single particles as they flow between arrays of microelectrodes. One set of electrodes is connected to an AC signal generator, whilst the opposite set are connected to current amplifiers. When a cell flows along the channel, it disturbs the AC current, and this change is measured as the impedance of that particle. Two frequencies (5 MHz and 40 MHz) are used, and in a typical experiment, around 20,000 single cells are measured in two minutes with a volumetric flow rate of 30 µL/min.

iFAST and BMD laboratory workflow

E. coli and *K. pneumoniae* strains (36 and 22, respectively) were streaked from –80°C glycerol stocks onto CLED plates and grown for 2 h at 37°C. A small amount of cells from this plate was used for MALDI TOF Biotyper speciation (Bruker MALDI Biotyper sirius System, MBT Compass HT software) using HCCA matrix (Bruker) to confirm that this was possible after short growth. A further small amount was added to 3 mL 0.9% saline and vortexed. 10 µL was added to 990 µL saline containing 5×10⁶ carboxylated polystyrene beads/mL, and the concentration of cells was established using the impedance cytometer at a flow rate of 30 µL/min for 30 s. As outlined in EUCAST guidelines, cell concentration was adjusted to 5×10⁶ cfu/mL in saline before adding 100 µL of the sample to 900 µL MH2 (final concentration 5×10⁵ cfu/mL) with (i) no antibiotics (control 1 and 2, first and last samples), (ii) antibiotics at breakpoint concentrations (see Table 1) or (iii) log₂ dilutions of antibiotics in MH2 in 96-well plates (BMD MIC). Samples i and ii were exposed for 2 hrs static at 37°C. Following exposure, 10 µL of each culture was diluted 1:100 in saline and measured as above. For the parallel BMD MIC (iii), plates were incubated at 37°C for 20 h and OD₆₀₀ read (BMG Labtech Clariostar). OD₆₀₀ 0.1 above blank was considered growth. The highest antibiotic concentration was 512 mg/L for all antibiotics except CIP (64 mg/L) and CAZ (128 mg/L). The concentration of potassium clavulanate for co-amoxiclav (AMC) was fixed at 2 mg/L.

Table 1
Enterobacterales EUCAST breakpoints (v12.0, 2022) of antibiotics used in this study (mg/L).

Antibiotic class	antibiotic	acronym	S ≤	I	R >	breakpoint conc. tested
β-lactam + inhibitor	co-amoxiclav	AMC	8/2	-	8/2	8/2
β-lactam	amoxicillin	AMX	8	-	8	8
cephalosporin (ESBL marker)	ceftazidime	CAZ	1	2 - 4	4	1
cephalosporin	cefalexin	CEX	16	-	16	16
fluoroquinolone	ciprofloxacin	CIP	0.25	0.5	0.5	0.25
aminoglycoside	gentamicin	GEN	2	-	2	2
nitrofurans	nitrofurantoin ^a	NIT	64	-	64	64
synthetic diaminopyrimidine agent	trimethoprim	TMP	4	-	4	4

^a Only for *E. coli* strains.

Clinical evaluation workflow

Urine confirmed to contain high *E. coli* count by growth on chromogenic agar were plated from 4°C onto CLED agar plates and incubated for 2–4 h at 37°C, depending on cell density, and confirmed as *E. coli* by MALDI TOF Biotyper speciation. The iFAST workflow was followed as above and compared to agar breakpoint tests as carried out by the diagnostic laboratory. To confirm results, the first three isolates (A, B and C) and nitrofurantoin for isolates G, H, I and J were repeated the next day from 4°C, after the agar breakpoint results were available,²⁹ in addition to disk diffusion assay, as carried out in parallel in the diagnostic laboratory.

Electronic MIC (eMIC) workflow

The laboratory workflow was followed as above, but bacteria were exposed to doubling dilution of antibiotics around the MIC for each strain in 1 mL MH2 and incubated at 37°C for two hours before diluting and reading on iFAST. A parallel BMD MIC from the same initial cell inoculum was conducted in 96 well plates according to EUCAST guidelines.

Whole genome sequencing analysis

Bacterial genomic DNA was extracted using the Promega Wizard DNA extraction kit according to the manufacturer's instructions. DNA was quantified by a Qubit fluorometer using the broad-range sensitivity dsDNA kit (Invitrogen). DNA was prepared and sequenced by the UK Health Security Agency Genomic Services and Development Unit (UKHSA-GSDU). Libraries were prepared using the Nextera DNA flex library prep kit according to the manufacturer's instructions, and reads (2×150 bp) were generated by the Illumina HiSeq 2500 platform. A minimum of 150 Mb of Q30 quality data was obtained for each sample. Fastq reads were quality trimmed using Trimmomatic v0.39 and draft chromosome contigs were assembled using SPAdes v3.15.3 filtering out contigs < 1 kb. Antibiotic resistance genes were identified using ResFinder.³⁰ Individual gene mutations leading to resistance were identified by manual analysis of known resistance genes, e.g. *gyrA*, in SeqBuilder (DNASTar Lasergene 14).

Data analysis and statistics

Impedance data was analysed using MATLAB (2023a), FlowJo (v10.10.0), Microsoft 365 Excel (v2208) and GraphPad Prism (v10.1.2). In FlowJo the 2 µm calibration beads added to each sample were used to normalise the impedance data for absolute cell volume (diameter 2, opacity 1, phase 0). Gates were manually drawn on the impedance scatter plots to identify electrical noise, beads and bacteria. Sample exclusion criteria for the bacterial gate were quality checked for the following features: a) control 1 (C1) < 1000 cells, b) bead count > 2-fold difference to C1, c) technical anomalies of read-out. The following metrics were determined: cell count, median and

coefficient of variance (CV) of diameter, opacity, and phase. For each metric the % of control 1 was calculated.

Analysed data was exported to Microsoft Excel and linked to MICs measured by BMD and sorted into susceptible and non-susceptible (resistant and susceptible at increased concentrations) per antibiotic as per EUCAST v12.³¹ *E. coli* and *K. pneumoniae* strains were treated equally except for nitrofurantoin, where *K. pneumoniae* strains were excluded from the analysis, as per EUCAST guidelines. A 2-tailed Wilcoxon matched-pairs signed rank test of control 1 vs control 2 was performed for all 7 metrics to confirm no significant difference (GraphPad Prism). Wilson/Brown ROC curves (95% CI) for each metric as % of control were drawn using susceptible/resistance data from BMD, with susceptible samples (by BMD) defined as negative and resistant samples positive. The best specificity and sensitivity to distinguish between susceptible and resistant for each antibiotic and metric were calculated using ROC curves (Fig. 3 and S3, Table S2 and S3). Thresholds for susceptibility were set at the value that had 100% sensitivity and 100% susceptibility as calculated by ROC. ROC curves were also generated excluding samples with MICs on or one doubling dilution above the breakpoint. Specificity of 100% reduces the chance of very major errors, i.e., situations where iFAST would call the strain susceptible, but MBD MIC would call it resistant. Statistical data is shown in Tables S2 and S3. Figures were created using Biorender.com.

Data statement

Raw data has been deposited in Mendeley Data DOI: 10.17632/47c4k2rfdj.1. Gene sequences have been deposited in BioProject Database under BioProject ID: PRJNA1246557.

Results and discussion

Developing the iFAST laboratory workflow

We have previously shown that an impedance-based fast anti-microbial susceptibility test (iFAST) can be used to determine the susceptibility of WHO priority pathogens to a range of antibiotics, each with different mechanisms of action.²⁰ This study extends this and examines the potential of the technology for rapid AST determination within a typical workflow as currently implemented for UTI ASTs in a large regional hospital in the UK.

The initial focus was on a panel of Enterobacterales, reflecting the most common pathogens found in UTI samples, namely *E. coli* and *K. pneumoniae*. The antibiotic panel is shown in Table 1 and consists of eight antibiotics widely used for frontline therapy of patients with a suspected UTI or for patient management in relation to carriage of isolates with extended spectrum beta lactamases (ESBLs) at University Hospital of Southampton (UHS) in 2022, in accordance with UK Standards for Microbiology Investigations²⁹ and UHS practice.

Fig. 1 compares the diagnostic laboratory workflow for anti-microbial susceptibility testing (AST) with iFAST and the clinical evaluation discussed later in this manuscript. The iFAST protocol

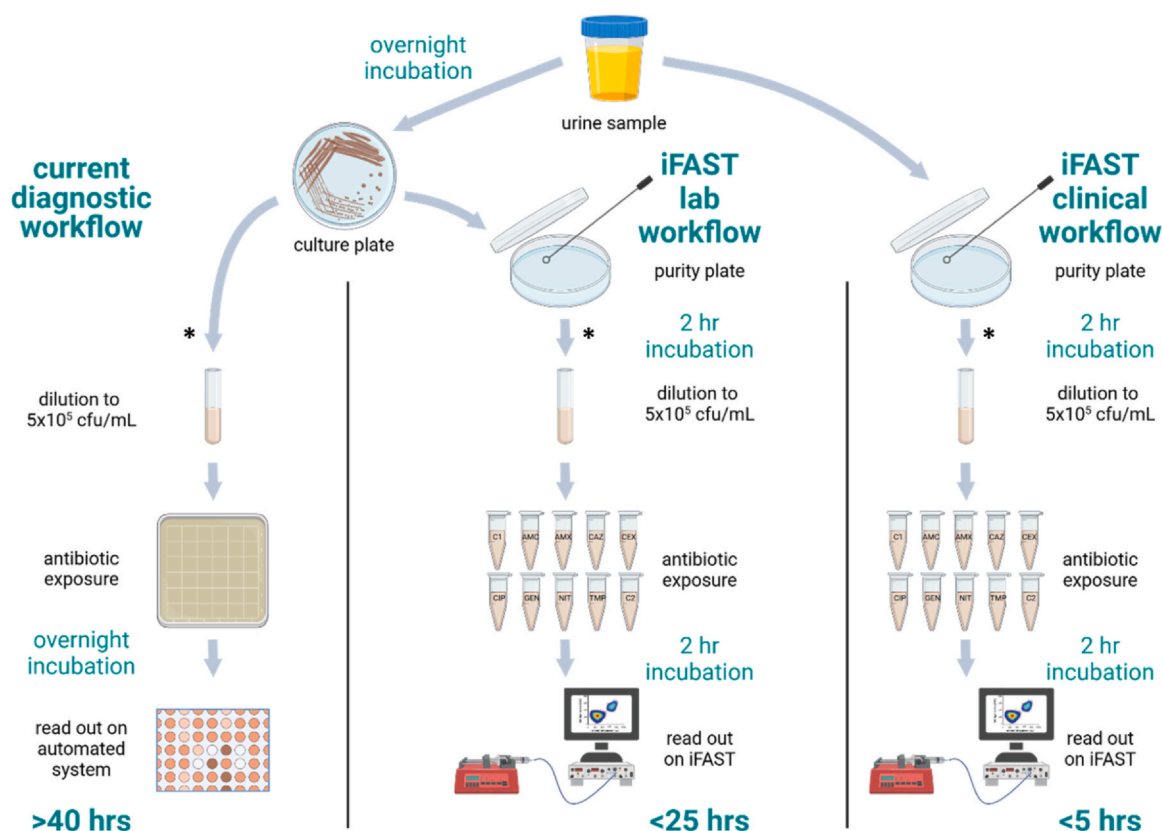


Fig. 1. Comparison of UTI AST diagnostic pathway as currently performed within a large regional hospital in the UK (left), the laboratory iFAST protocol (centre) and the clinical evaluation iFAST protocol (right). The current diagnostic workflow requires two overnight incubations from the urine sample. First to isolate colonies on CLED agar, second to assess susceptibility to the antibiotic. The iFAST lab workflow retains the initial plating step. To have logarithmically growing cells, isolates are restreaked on a second agar plate for two hours before exposure to antibiotics for two hours, rather than overnight as in the diagnostic workflow. The clinical evaluation workflow only has the 2x two hour incubations from the urine sample to iFAST readout. This means that the complete iFAST workflow can be carried out in less than five hours, direct from a positive urine sample, and thus in less than one working shift (eight hours). *identification using MALDI was carried out at this point.

requires plating for two hours to produce logarithmically growing cells (Fig. S2), followed by exposure to antibiotics for a further two hours, instead of overnight incubation as for the current diagnostic workflow, before reading the sample on iFAST (2 min per sample). To closely align the method with EUCAST/ CLSI MIC testing, cells were diluted to a concentration of 5×10^5 cfu/mL in MH2. Each antibiotic in the panel of eight was set to the concentration of the susceptibility breakpoint for *Enterobacterales* as in EUCAST published values (Tables 1, v12.0, 2022). It has been previously shown that the effect of a beta-lactam antibiotic can be observed by impedance cytometry within 30 min.³² To accommodate the mode of action of the range of different antibiotics, all samples were exposed for two hours. The final protocol is described in Fig. S1.

The panel of *E. coli* and *K. pneumoniae* strains (36 and 22, respectively) is shown in Table S1, including their antibiogram based on BMD MIC testing and resistance mutations. Each strain was exposed to breakpoint concentrations of each antibiotic, together with two un-exposed controls for two hours followed by resuspension in saline for measurement on the impedance cytometer (measurement taking two minutes per sample). Each test was carried out once, unless technical quality control metrics were unmet (see Materials and methods), in which case parallel iFAST and BMD were repeated. The MICs determined by BMD were used as a direct comparator for the breakpoint measurements using the impedance cytometer.

Example impedance scatter plots are shown in Fig. 2a. The x-axis is the electrical diameter of the cells (determined from the impedance at 5 MHz) and the y-axis the electrical opacity, a ratio of high frequency to low frequency impedance, or electrical phase (high frequency only), which reflects different electrical phenotypic

changes in the cell membrane. The scatter plots show the cell population along with the beads that are used for normalisation. Fig. 2b shows an example data set for *E. coli* and *K. pneumoniae* exposed to the panel of antibiotics at the clinical breakpoint. Each figure in the panel shows an impedance scatter plot for 60 μ L of control or exposed samples. The scatter plots clearly show strains that were susceptible to antibiotics, where the number of cells decreased compared to control, or resistant, where there was negligible change in the cell count and position. Strain NCTC 13476 was resistant to all antibiotics except NIT. The other four strains were susceptible to some or all of the antibiotics. Similar results were obtained for other *E. coli* and *K. pneumoniae* strains (note that the EUCAST-defined breakpoints are the same for both species, except for nitrofurantoin, which is recommended for use with *E. coli* only).

Susceptible vs resistant metrics

Seven different metrics were used to compare iFAST data with the BMD results, namely: (i) cell count, (ii) coefficient of variance (CV) of diameter, (iii) CV of opacity, (iv) CV of phase, (v) median diameter, (vi) median opacity and (vii) median phase. In each case the % of exposed compared to unexposed control was determined. Two unexposed control samples were routinely used to identify any potential errors. No significant differences were observed between the two controls for any of the seven metrics for any of the samples, demonstrating that the technical variability of the test is negligible.

BMD assays have an inherent 2-fold variability (i.e. an MIC of 8 mg/L may in fact be 4 mg/L or 16 mg/L). This is recognised as essential agreement in ISO 20776-1:2019.²¹ In this work we are

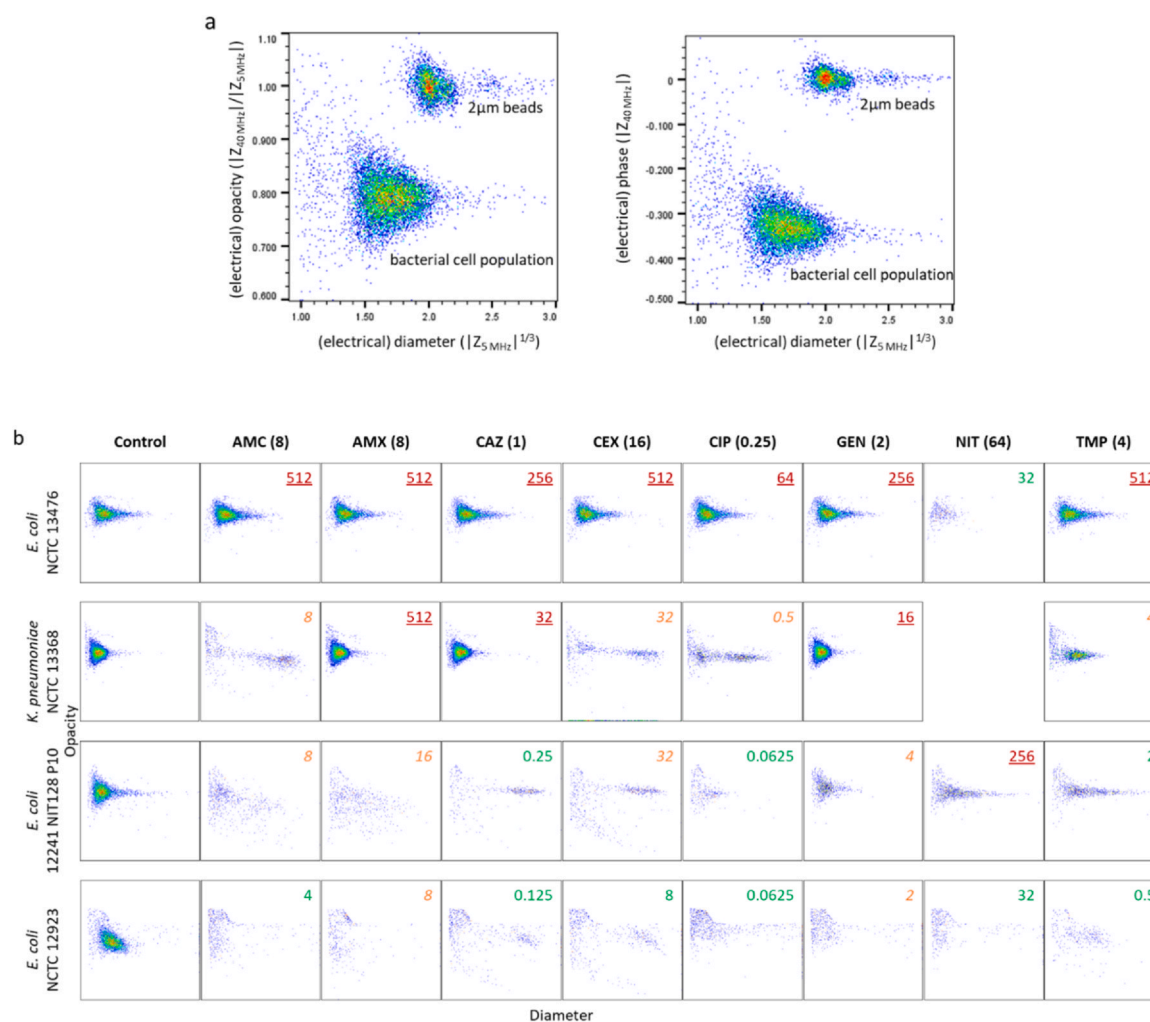


Fig. 2. Electrical changes in cell populations exposed to breakpoint concentrations of eight antibiotics can define the susceptibility of strains. **a** Example impedance scatter plot of unexposed control *K. pneumoniae* NCTC 13368 together with 2 μm diameter polystyrene beads (used for normalisation). The x-axis is the cube root of the impedance (proportional to diameter) measured at a frequency of 5 MHz. The y-axis is the electrical opacity, the ratio of high frequency (40 MHz) to low frequency (5 MHz) impedance (left image) or the phase, the high frequency (40 MHz) not normalised to cell size (right image), which measure the electrical properties of the cell membrane. **b** Scatter plots of cell populations exposed to breakpoint concentrations of eight antibiotics (mg/L) for two hours measured by iFAST and standard BMD MIC (mg/L, in top right of each scatterplot, green, plain = MIC below breakpoint, orange, italic = MIC equals breakpoint concentration or one above, red, underlined = MIC above breakpoint). Readouts were normalised to the beads, and electrical noise and beads gated out of the plots shown. NIT is not applied to *K. pneumoniae* strains as per EUCAST guidelines.

comparing a quantitative test (MIC) to a qualitative test (S/R). In addition, the number of samples within the two-fold variability of the “gold standard” BMD test varied between antibiotics. In order to account for these factors, we excluded for each antibiotic, those strains that had a BMD MIC on, or one doubling dilution above, the susceptibility breakpoint tested.

The optimal metric that could distinguish between susceptible and resistant for each antibiotic was determined using Receiver Operating Characteristic (ROC) curves (Fig. 3, Table S2).

All eight antibiotics showed a concordance of 100% specificity and 100% sensitivity between BMD MIC test and iFAST breakpoint exposure when cell count was used as a metric to discriminate susceptible and resistant isolates (Fig. 3, Table S2). Due to the MICs of the strains included in the analysis, only 7 and 20 strains were susceptible and 37 and 6 resistant to AMX and NIT, respectively, reducing the confidence in the susceptibility threshold values for distinguishing susceptible strains for these antibiotics. The only non-concordances for all antibiotics were for strains that had a BMD MIC within the variability of the gold standard, i.e., an MIC on or one doubling dilution above the breakpoint concentration. The only exceptions to this were two strains in which plasmid loss had occurred (see supplementary information).

When all data, including those with an MIC at or one doubling dilution above the breakpoint, were analysed, the ability of iFAST to distinguish between susceptible and resistant strains was high with sensitivity between 100% (CAZ) and 80% (CEX) and an outlier of 30.77% (NIT) at specificity of 100% (Fig. S3, Table S3).

Given the above limitations of strains with an MIC on or one doubling dilution above the susceptibility breakpoint, concordance between the iFAST breakpoint test and BMD MIC was 100% for all eight antibiotics tested against the 58 *E. coli* and/or *K. pneumoniae* isolates (Fig. 3, Table S2).

iFAST clinical workflow evaluation

The applicability of iFAST in a clinical setting was evaluated in a pilot study of 10 urines with heavy *E. coli* growth following the protocol outlined in Fig. 1. These were tested against the same panel of eight antibiotics alongside agar breakpoint tests, as carried out in the diagnostic laboratory at the time. As expected, most isolates were susceptible to all or most of the antibiotics tested. Using the same quality control and threshold parameters of % cell count compared to control determined from the 58 strains, it was established that the susceptibility results determined by iFAST matched

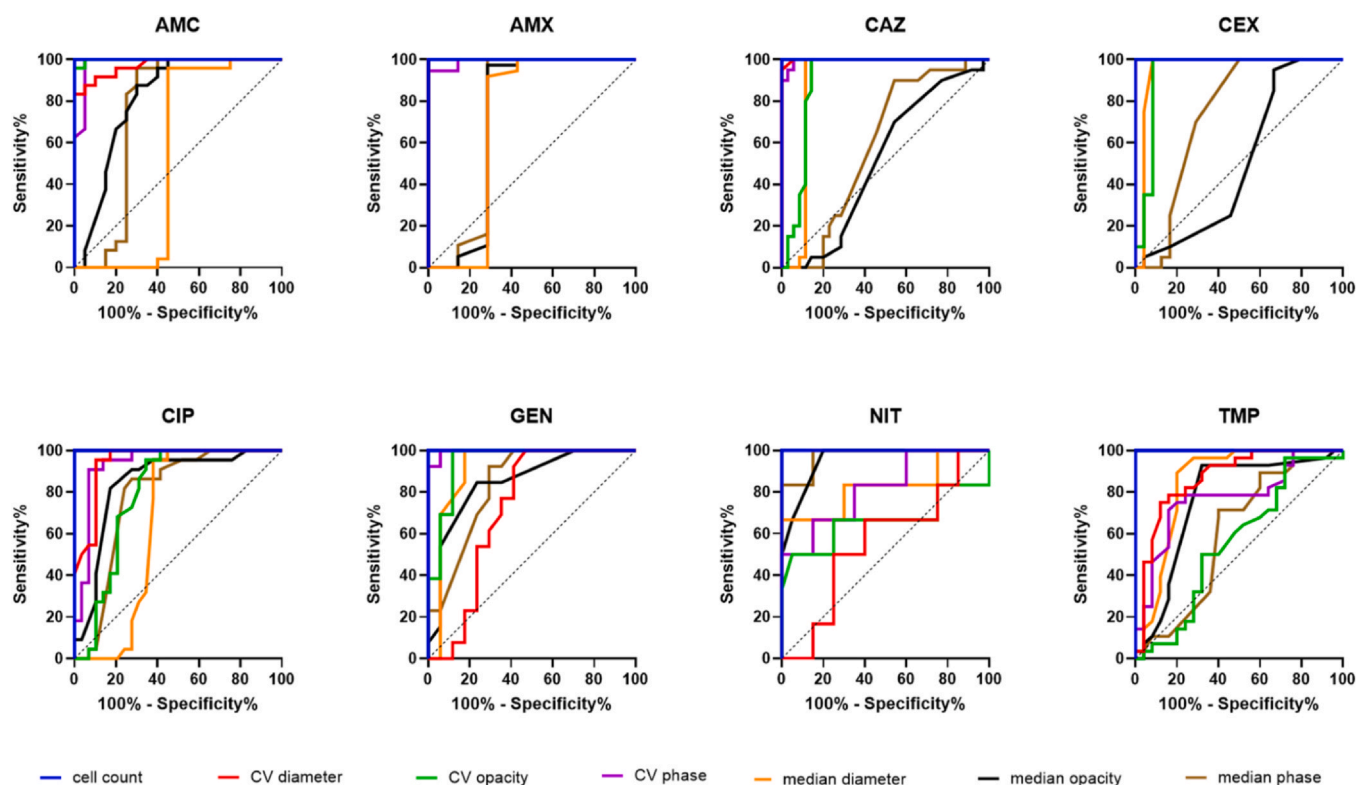


Fig. 3. ROC curves comparing iFAST with BMD demonstrating 100% concordance for all eight antibiotics. In these plots, samples with an MIC in the two-fold variability for the gold standard broth microdilution AST method have been removed. In all cases the % cell count compared to unexposed control (blue line) metric could discriminate susceptible from resistant strains when compared to the gold standard. Dotted line represents a random guess. For NIT only *E. coli* strains were used for the analysis.

the laboratory diagnostic test in 74 out of 80 tests, as summarised in Table 2 and Table S4.

Two mismatches between iFAST and agar susceptibility testing were initially found each for co-amoxiclav, amoxicillin and trimethoprim and seven mismatches for nitrofurantoin (Table S4). Five of the mismatches were subsequently retested using disk diffusion (isolates A, B and C for all antibiotics), with three of the new results matching the original iFAST result (green shading in Table 2), highlighting variability in the agar AST method. The thresholds to distinguish between susceptible and resistant are low for AMC (17% of control) and NIT (15%) compared to the other antibiotics (between 37% and 47%). This means that these two antibiotics are more prone to cell debris in the scatterplot outweighing live cells when overall cell counts are low, thereby distorting the result. In fact, isolates D, E, F, G, H, I and J, which had

mismatches for nitrofurantoin, had control cell counts below 2000 and isolate E had cell counts below quality control threshold for the control (<1000, supp Table 4). In a repeat with higher cell count, NIT iFAST matched the agar susceptibility profile in all tested cases (supp Table 4).

Only one of the remaining five mismatches (Isolate B AMC and NIT, Isolate D AMC, NIT and TMP, red shading in Table 2) would be classified as a very major error, i.e., calling an isolate susceptible when it is resistant. This isolate (isolate B AMC in Table 2) falls below the susceptibility threshold for co-amoxiclav (12% < 17% cell count of control). When isolate B was exposed to increasing concentrations of AMC and analysed on iFAST, it became clear that this isolate has an MIC on the breakpoint (Fig. S4b). Once again, this shows that iFAST had essential agreement with the diagnostic method used, explaining the mismatch.

Table 2

Final susceptibility data for clinical urine *E. coli* isolates comparing the iFAST method with agar AST and disk diffusion.

Isolate	A			B			C			D		E		F		G		H		I		J	
	iFAST	agar	DD	iFAST	agar	DD	iFAST	agar	DD	iFAST	agar	iFAST	agar	iFAST	agar	iFAST	agar	iFAST	agar	iFAST	agar	iFAST	agar
co-amoxiclav	S	S	S	S	R	R	S	S	S	R	S	S	S	S	S	S	S	S	S	S	S	S	S
amoxicillin	S	R	S	R	R	R	S	R	S	R	R	S	S	R	R	R	R	R	R	R	R	S	S
ceftazidime	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
cephalexin	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
ciprofloxacin	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
gentamicin	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
nitrofurantoin	S	S	S	R	S	S	S	S	S	R	S	R	S	S	S	S	S	S	S	S	S	R	S
trimethoprim	S	S	S	R	S	R	S	S	S	R	S	S	S	R	R	R	R	S	S	S	S	S	S

S: Susceptible, R: Resistant, agar: agar AST in diagnostic lab, DD: disk diffusion AST in diagnostic lab. Red shading: non-concordance between iFAST and diagnostic lab, green shading: non-concordance between iFAST and diagnostic lab result where iFAST result was confirmed using a secondary confirmatory method (DD), yellow shading: iFAST initially read resistant, repeat with higher cell count confirmed susceptible, greyed out: cell count of control below quality control threshold. The threshold values in Fig. 3, Table S2 were used to define S/R using % cell count compared to control for all antibiotics (see Table S4).

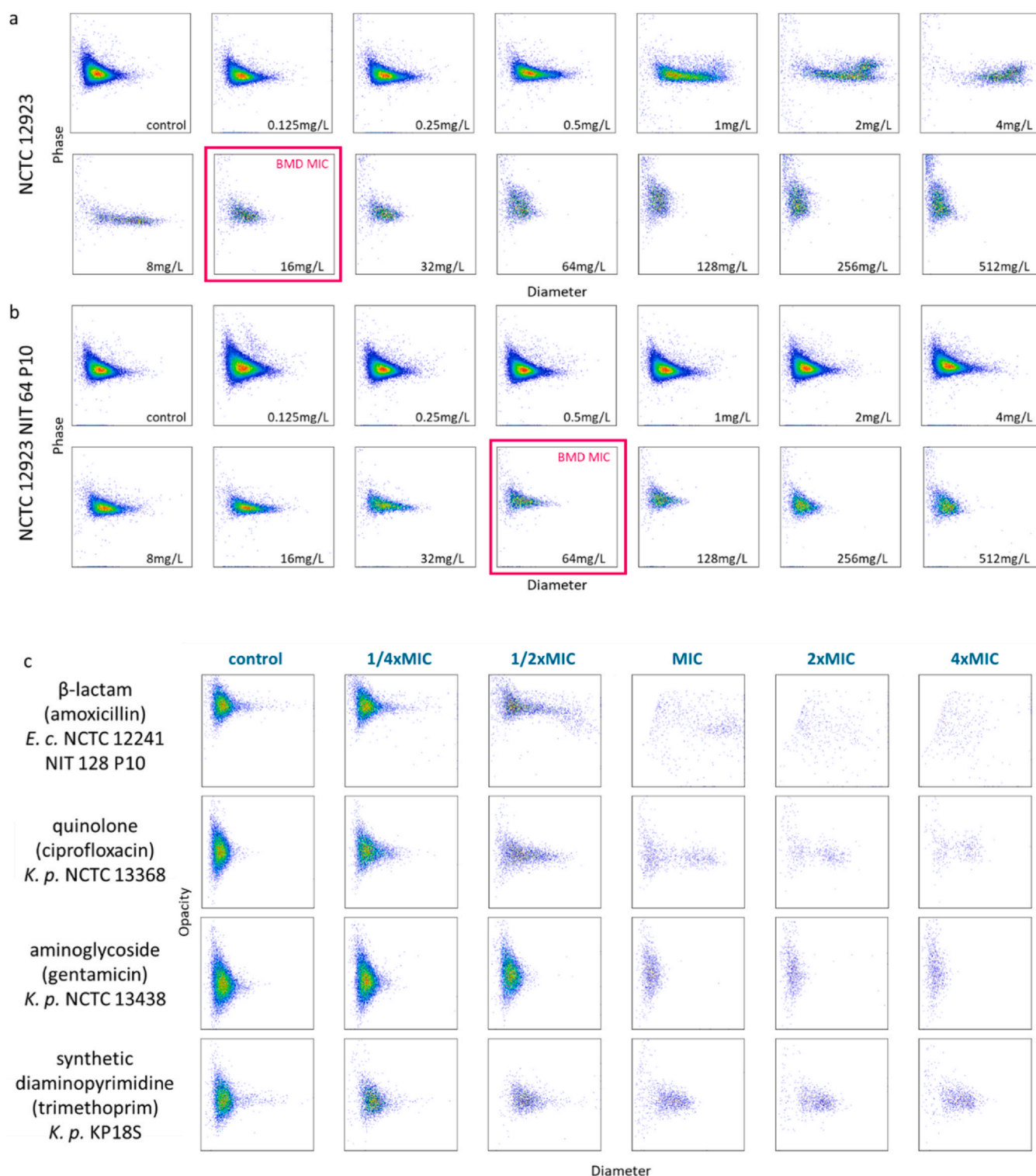


Fig. 4. Bacterial cells change their electrical properties at sub-MIC levels. Series of impedance scatter plots for cells exposed to doubling dilutions around the MIC of various antibiotic classes (eMIC). **a** NCTC 12923: change in cell count, diameter and phase exposed to sub-MIC concentrations of NIT **b** 12923 NIT64 P10: decreased cell count, no change in diameter or phase at sub-MIC concentrations. **c** Scatter plots showing different responses to a range of antibiotic classes at sub- and supra-MIC levels. These changes are indicative of biophysical changes of the bacteria upon exposure to the antibiotic, e.g., stress responses, permeability changes or membrane changes. BMD MIC: MIC as determined by the parallel broth micro dilution.

Isolate D exposed to co-amoxiclav and trimethoprim had populations above the respective thresholds (28% and 53%, thresholds 17% and 40% respectively), determining them resistant. This isolate was not retested with higher cell concentrations nor using disk diffusion, so either result may have been correct (Fig. S4a).

To explain the mismatch of NIT in isolates B and D, further understanding of the impact of nitrofurantoin on the electrical properties of cells after only two hours of exposure would help to establish a more reliable predictive threshold for this antibiotic. For all other antibiotics the established thresholds were clearly able to identify susceptible *E. coli* isolates.

Electrical MIC (eMIC)

Comparison of BMD MIC to a quantitative electrical minimum inhibitory concentration using the iFAST system (eMIC) could allow reliable differentiation between nitrofurantoin-resistant and susceptible isolates. We therefore measured susceptibility at more than one antibiotic concentration on iFAST for nine laboratory strains with varying nitrofurantoin MICs (Table S5). Example scatter plots are shown in Fig. 4. Results showed that the response to antibiotics varied between strains, especially at exposure to sub-inhibitory concentrations. Susceptible strain NCTC 12923 had changes in cell count, diameter and phase when exposed to a sub-MIC antibiotic concentration (Fig. 4a). In contrast, NCTC 12923 NIT 64 P10, which is the same strain adapted to NIT (mutations in NfsA and NfsB), showed no change in diameter and phase following exposure to sub-MIC concentrations of NIT, but had a reduced cell count (Fig. 4b). At the MIC, the cell count did not decrease further, but the diameter decreased to below that of the unexposed cells. Comparing NIT susceptibility as measured by BMD MIC and eMIC (% cell count of control) showed essential agreement in five out of nine cases. However, when the second-best metric, % change in median phase, was chosen (see supp Table 2 and Fig. 3), nine out of nine NIT eMICs showed essential agreement with the BMD MIC. NIT concordance may therefore be improved by performing eMICs and more clearly defining the best metric in more samples, rather than only break-point testing and defining susceptibility by reduction in cell count.

Further eMICs were performed on several mismatched strain/antibiotic combinations to investigate whether concordance would improve for strains with an MIC on or one doubling dilution above the breakpoint, as for isolate 2 in the clinical evaluation (Fig. S4, Table S5). Though only 60% showed essential agreement with the BMD MIC, the mismatches always showed a lower eMIC than BMD MIC. This may be due to changes in the electrical properties of exposed cells being detectable at sub-MIC levels, depending on antibiotic (Fig. 4c). In particular for the β -lactam (amoxicillin) and the fluoroquinolone (ciprofloxacin) a significant decrease in event count together with an increase in electrical diameter was observed at sub-MIC levels. Cell count further decreased at the MIC, and electrical diameter decreased to below that of the unexposed cells, as also seen for nitrofurantoin. For the aminoglycoside gentamicin the cell count was reduced but there was no change in any other metrics. Results for trimethoprim were similar to gentamicin, with a small increase in diameter at sub- and supra-MIC. These data demonstrate that iFAST can identify subtle changes in populations following antibiotic exposure, which might further our understanding of resistance and susceptibility of individual strains to different antibiotic classes, as well as modes of action of antibiotics or mechanisms of resistance to them. This complexity makes it difficult to determine a single metric for interpretation of eMICs against all antibiotics.

Conclusions

This work has demonstrated that a new single-cell impedance cytometric AST (iFAST) can be developed to deliver reliable susceptibility results more than 35 h faster than the standard AST with comparable accuracy for *E. coli* and *K. pneumoniae* against eight commonly used antibiotics. As with the current gold standard BMD method, the only strains that were miscalled susceptible or resistant were those with an MIC on or one doubling dilution above the MIC, meaning that there was in all cases essential agreement between the BMD and iFAST read-outs. Exposure to more than one concentration, such as in eMICs, gives a more accurate susceptibility read-out.

The current gold standard BMD assay is based on measurement of increases in bacterial cell count, usually assessed by eye. Other fast phenotypic ASTs under development measure rate of cell growth or other single surrogate metrics such as nanomotion¹⁴ or viability

markers.¹³ iFAST measures changes in cell growth, but also other metrics, such as changes in cell size, permittivity and conductivity, allowing for a more accurate picture of the cells' responses to antibiotic exposure at the population level. As iFAST measures the electrical changes in single live cells, no chemical fixing, cross-linking or expensive dyes are required. The only consumables required are saline, beads, media, antibiotics and commonly available 96-well plates. In addition, no complex optical systems are used, allowing iFAST to be more easily automated and accelerated in both speed and throughput. Although commercial decisions are outside of the remit of this study, this might allow the technology to be available at a price comparable to current disc diffusion AST.

Future work will concentrate on expanding the clinical applicability of the technology with a larger concordance study on clinical UTI samples, increasing the number of bacterial species, including more ESBL phenotypes, antibiotics and antibiotic concentrations, together with optimising nitrofurantoin and other antibiotic read-outs. We will further test the impact of samples that may contain more than one species (e.g. *Staphylococcus aureus* contamination on collection) to develop a more reliable method, which works directly from urine. Implementation of a rapid test would contribute to antibiotic stewardship and speed up the diagnostic process whilst reducing hands-on time for staff.

Declaration of Competing Interest

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hywel Morgan and Daniel Spencer own equity in a spin-out company iFAST Diagnostics Ltd.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jinf.2025.106549.

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